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Rankuno AI Engine - Architectural Blueprint

Master SDLC Standards, Agentic Architecture, & Phase 1 Page Classification Engine Specification

1. Executive Summary & Core Architecture Stack

The **Rankuno AI Automation Infrastructure** is built upon the **Google Antigravity Agentic Stack**. It establishes a 0-to-1 engineering framework for automating SEO operations, Paid Search (PPC), and Market Research with zero tech debt.

Layer	Component	Google Antigravity Equivalent	Rankuno Implementation Role
Model	Reasoning Engine	Gemini 3.6 Flash / Pro	Reasoning, architectural planning, and code generation.
Agent	Decision Maker	Antigravity Execution Loop	Primary orchestrator executing Plan → Act → Observe → Reason loop.
Subagents	Specialized Delegates	<code>invoke_subagent</code>	Background workers for research, audits, or isolated test suites.
Skills	Procedural Guides	<code>skills/<name>/SKILL.md</code>	Reusable procedural guides (<code>antigravity-sdlc-flow</code> , <code>seo-engine-guide</code>).
Tools	Native Capabilities	Atomic capabilities	<code>read_file</code> , <code>write_file</code> , <code>run_command</code> , <code>grep_search</code> .
MCP	External Connectors	Model Context Protocol	Connectors for Search Console, Google Ads API, SERP scrapers.

2. Mandatory 8-Step SDLC Execution Protocol

Every feature, module, or update in this repository MUST follow the strict 8-step engineering pipeline:

- Investigation & Requirement Discovery:** Inspect codebase, verify domain rules and existing structures.
- System Architecture & Data Schema Blueprint:** Define Pydantic contracts and component boundaries.
- Human-in-the-Loop (HITL) Review Checkpoint:** Present architecture to operator for explicit approval.
- Reasoning-Driven Implementation Plan:** Detail line-by-line file changes and logic.

- 5. **Security, Rate-Limit & Edge-Case Audit:** Audit for anti-bot blocks, rate limits, API spend, breaking changes.
- 6. **Step-by-Step Modular Implementation:** Write clean, strictly typed, modular code.
- 7. **Automated Unit Testing & Verification:** Execute unit test suites (pytest); never declare success without passing tests.
- 8. **README & Architecture Drift Audit:** Keep documentation in sync with codebase changes immediately.

3. Phase 1: Multi-Dimensional Page Classification Engine

Phase 1 classifies website URLs across **3 complementary interfaces** to build a structured content graph:

- **Interface 1: Page Type & Hierarchy** (L0 Homepage , L1 Primary Nav Hub , L2 Sub-Nav Hub , L3 Leaf Page , Utility).
- **Interface 2: Theme & Topical Cluster** (Business Niche, Content Silos, Subject Matter).
- **Interface 3: Semantic Intent & Conversion Role** (Informational, Commercial Investigation, Transactional, Navigational).

3.1 The 6-Signal Consensus Architecture (Fail-Safe Strategy)

To guarantee zero classification failures on complex or flat websites, the engine evaluates 6 independent signals:

```
1. ARIA Nav Tree Signal --> Inspects Accessibility ARIA tags (role="navigation") & nested <ul>/<li> (Resolv
2. CMS API Endpoint    --> Queries WordPress (/wp-json/) & Shopify (/collections.json) endpoints (Resolves
3. Grouped XML Sitemaps --> Parses product-sitemap.xml, category-sitemap.xml, post-sitemap.xml.
4. Schema.org JSON-LD  --> Parses @graph types (CollectionPage, Product, Article, Service).
5. Link In-Degree      --> Evaluates internal link weight (>=1000 links = L1 Primary Hub).
6. LLM Zero-Shot Backup --> Gemini 3.6 Flash fallback when structural signals are ambiguous.
```

3.2 Pydantic Data Contracts

```
from enum import Enum
from typing import List, Optional
from pydantic import BaseModel, Field

class HierarchyLevel(str, Enum):
    L0_HOMEPAGE = "L0_HOMEPAGE"
    L1_PRIMARY_NAV_HUB = "L1_PRIMARY_NAV_HUB"
    L2_SUB_NAV_HUB = "L2_SUB_NAV_HUB"
    L3_LEAF_PAGE = "L3_LEAF_PAGE"
    UTILITY_PAGE = "UTILITY_PAGE"

class PrimaryPageType(str, Enum):
    HOMEPAGE = "HOMEPAGE"
    SERVICE_CATEGORY_HUB = "SERVICE_CATEGORY_HUB"
    SERVICE_DETAIL_PAGE = "SERVICE_DETAIL_PAGE"
    PRODUCT_CATEGORY_HUB = "PRODUCT_CATEGORY_HUB"
    PRODUCT_DETAIL_PAGE = "PRODUCT_DETAIL_PAGE"
    BLOG_HUB = "BLOG_HUB"
    BLOG_ARTICLE = "BLOG_ARTICLE"
    COMPANY_ABOUT = "COMPANY_ABOUT"
    COMMERCIAL_LEAD_GEN = "COMMERCIAL_LEAD_GEN"
    FACETED_FILTER = "FACETED_FILTER"
    UTILITY_LEGAL = "UTILITY_LEGAL"
    UNKNOWN = "UNKNOWN"

class SignalScore(BaseModel):
    source: str
    suggested_level: HierarchyLevel
    suggested_page_type: PrimaryPageType
    confidence: float = Field(ge=0.0, le=1.0)
    notes: str

class FullPageIntelligenceProfile(BaseModel):
    url: str
    normalized_path: str
    hierarchy_level: HierarchyLevel
    primary_page_type: PrimaryPageType
    nav_parent_url: Optional[str]
    breadcrumb_path: List[str]
    topical_category: str
    sub_topic: Optional[str]
    search_intent: str
    conversion_role: str
    signals_evaluated: List[SignalScore]
    final_confidence_score: float = Field(ge=0.0, le=1.0)
    consensus_method: str
```

4. Edge-Case & Security Audit Matrix

Edge Case / Threat	Vulnerability / Cause	Rankuno Architectural Guardrail
Hamburger / Hidden Mobile Navs	CSS hides menu (<code>display:none</code>).	Inspect DOM ARIA tree (<code>role="navigation"</code>) ignoring visual CSS.
Flat URLs (<code>site.com/page</code>)	Directory path regex breaks.	Query CMS Endpoints (<code>/collections.json</code>) & Sitemap index grouping.
JavaScript SPAs	Empty <code><div id="root"></code> returned.	Playwright headless browser rendering fallback.
Multi-Language Paths	Locale prefixes skew paths (<code>/en/services</code>).	Path Normalizer strips locale prefixes before hierarchy matching.
Faceted Filter Params	Query params (<code>?color=red</code>) create infinite URLs.	Parameter Normalizer flags parameterized URLs as <code>FACETED_FILTER</code> .
API Cost Overflow	Unbounded LLM calls.	Rate-limiter & caching layer: LLM invoked only when 5 structural signals are ambiguous.